

Project

Before your start kindly visit the link below it will guide you with some python packages that are very simple but very effective and can help you with below projects and also help you out in long run. There are in total 3 ideas arranged in increasing difficulty.

[Google Colab Link](#)

Note: For all the ideas below try to implement them with using learnings from functions

Idea 1 (EASY)

Try to create a guessing game, where a dice will be rolled 5 times and the player needs to guess the number. If the player can guess the number correctly then the player gets 1 point. Player will win when 3 point is reached otherwise the ai or computer will win.

Idea 2 (Medium)

Create an expense tracker that will keep record of money that you earned and spend. For expense there will be four categories food, travel, friends and others. For income there will be 3 categories pocket money, job and others. This looks simple right? Don't worry there is more things to come. You should create a function that can generate a report whenever it is called, this report will show what expenses and income you made in each category and finally total expense and result.

Let me help you out with what functions do you need for this:

1. **add_expense(category, amount)**
2. **add_income(category, amount)**
3. **generate_report()**

You are most welcome to add more features to it but the above 3 features are must

Idea 3 (Hard)

I am pretty sure everybody watched Squid Game and the marble game was the best game in my opinion, therefore let's try to recreate that game with python. And btw if you did not watch the series I'll explain the game below so that you can get your hand dirty with python code.

There will be two players (One you and another AI), both will have 10 marbles to start with. This game will be in turns. There will be a coin toss which will always be called by the player. If the player wins then he or she can choose whether to guess first or not. If the player chooses to guess first then the AI will make a bet of any amount of marbles that it has. After the bet is done the player will guess how much marbles did the AI bet. If the player is correct then AI will lose that amount of marble that he betted, on otherhand if the player makes a wrong guess then the player needs to give the amount of marble that is betted by the AI. In next turn the player will bet and the AI will guess. In this way the game will continue until one of the player don't have any marbles.

Constraints:

Each player can bet only the amount of marble he or she has. For example if a player has only 3 marbles then he or she can only bet maximum 3 marbles not more than that, but he or she is welcome to bet less than 3 but greater than 0.

If you want to make this game more interesting then you are welcome to add twists to it by adding new rules.